

**Daniel Brouwer (Attempt****1 of 1)**

Test Information

Current Grade 70.00000 out of 100 points

Grade based on Last Evaluated Attempt

Status Completed

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Due Date 12/1/25 11:59 PM

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Instructions

QUESTION 1: CALCULATED NUMERIC

0

out of 25 points

Refer to page 334 of Chapter 7 that is uploaded on the UIC Blackboard in the Course Textbook content area.

1. Identify the error(s) in the calculation of the Dew Point pressure
2. Propose a detailed algorithm for the correct calculation of the Dew Point pressure.
3. Carry out the calculation and enter your result in **bar** in the textbox below. Round your numerical answer into **two decimal digits**.



Given Answer: 2.56

Correct Answer: 3.7

Answer range +/- 0 (3.7 - 3.7)

QUESTION 2: FILE RESPONSE

70

out of 75 points

Midterm 3: Dew–Point Pressure Calculation for an Ethanol–Toluene Mixture

Verification and Correction of Example 7.2 (Chapter 7)

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1 Introduction

The purpose of this report is to analyze, verify, and correct the dew–point pressure calculation presented in Example 7.2 of Chapter 7 (“The Method of Liquid Activity Coefficients”). The example considers a binary mixture of ethanol (component 1) and toluene (component 2) at a fixed temperature of $T = 150\text{ }^{\circ}\text{C}$ and overall composition $(z_1, z_2) = (0.30, 0.70)$. The system pressure is 5 bar, and the goal is to show that the state is two–phase and to perform a flash calculation.

The Chapter 7 text reports a bubble–point pressure of $P_{\text{bubble}} = 6.6965\text{ bar}$ and a dew–point pressure of $P_{\text{dew}} = 3.948\text{ bar}$. In this report we focus on the dew–point calculation. We (i) identify the errors in the published dew–point result, (ii) derive a correct dew–point algorithm based on the Rachford–Rice equation and activity–coefficient models, and (iii) recompute the dew–point pressure using the data provided in the example.

2 Problem Statement and Given Data

For the ethanol–toluene mixture at $T = 150\text{ }^{\circ}\text{C}$ the example gives the following quantities:

- Overall (feed) composition:

$$z_1 = 0.30 \quad (\text{ethanol}), \quad z_2 = 0.70 \quad (\text{toluene}).$$

- Liquid activity coefficients at the stated temperature and composition, from the Wilson model via the CHE Calculator:

$$\gamma_1 = 1.855753, \quad \gamma_2 = 1.107145.$$

- Saturated vapor pressures at the same temperature:

$$P_1^{\text{sat}} = 8.2626 \text{ bar}, \quad P_2^{\text{sat}} = 1.7051 \text{ bar}.$$

These data are used by the example both for the bubble- and dew-point calculations.

3 Theoretical Background

3.1 K-values and activity coefficients

For VLE at temperature T and pressure P , with non-ideal liquid behavior and (approximately) ideal vapor behavior at low to moderate pressures, the phase equilibrium condition for component i is

$$y_i P = x_i \gamma_i P_i^{\text{sat}}(T), \quad (1)$$

where x_i and y_i are the liquid and vapor mole fractions, γ_i is the liquid activity coefficient, and $P_i^{\text{sat}}(T)$ is the pure-component saturated vapor pressure. Defining the K-value

$$K_i = \frac{y_i}{x_i}, \quad (2)$$

we obtain the familiar expression

$$K_i = \frac{\gamma_i P_i^{\text{sat}}}{P}. \quad (3)$$

3.2 Rachford–Rice equation and dew-point condition

For a flash calculation at fixed T , P , and overall composition z_i , the Rachford–Rice equation is

$$F(\alpha) = \sum_i \frac{z_i (K_i - 1)}{1 + \alpha (K_i - 1)} = 0, \quad (4)$$

where α is the vapor fraction.

- For a *bubble point* at given T , $\alpha = 0$ and (4) reduces to $\sum_i z_i K_i = 1$.

- For a *dew point* at given T , $\alpha = 1$ and (4) reduces to

$$\sum_i \frac{z_i}{K_i} = 1. \quad (5)$$

Substituting the K -value expression (3) into the dew-point condition (5),

$$\sum_i \frac{z_i}{K_i} = \sum_i \frac{z_i P}{\gamma_i P_i^{\text{sat}}} = P \sum_i \frac{z_i}{\gamma_i P_i^{\text{sat}}} = 1, \quad (6)$$

we obtain a compact formula for the dew-point pressure at fixed temperature and composition:

$$P_{\text{dew}} = \left[\sum_i \frac{z_i}{\gamma_i P_i^{\text{sat}}(T)} \right]^{-1}. \quad (7)$$

Equation (7) is valid when the activity coefficients γ_i are evaluated for the liquid composition that is in equilibrium with the specified vapor composition $\{z_i\}$. In practice, this requires an iterative calculation because the liquid composition is not known a priori.

4 Identification of Errors in the Example

The text of Example 7.2 proceeds as follows:

1. In **Step 3**, the example sets $\alpha = 1$ in the Rachford–Rice equation and derives a dew-point relationship that is consistent with (5).
2. In **Step 4**, the example evaluates the dew-point pressure numerically using the given values of γ_i , P_i^{sat} , and z_i , and reports

$$P_{\text{dew}} = 3.948 \text{ bar}.$$

Recalculating with the given numbers (see Section 6) shows that Eq. (7) yields $P_{\text{dew}} \approx 2.56 \text{ bar}$, not 3.95 bar. Therefore there is at least one numerical inconsistency in the published dew-point value.

In addition, the example reuses the activity coefficients γ_i that were computed at the overall composition $x_i = z_i$. For a strictly correct dew calculation, however, γ_i should be evaluated at the *equilibrium liquid composition* corresponding to the specified vapor composition and dew-point pressure. This requires an iterative procedure, which is not carried out in the example. Thus we can identify two errors:

1. **Numerical / implementation error:** the reported value $P_{\text{dew}} = 3.948$ bar is inconsistent with the stated data and the dew-point equation. Direct substitution of the given γ_i , P_i^{sat} , and z_i into Eq. (7) gives $P_{\text{dew}} \approx 2.56$ bar.
2. **Algorithmic / conceptual error:** the example does not iterate on the liquid composition when computing the dew point. Instead, it uses γ_i evaluated at the overall composition. A fully consistent dew-point calculation must update both the liquid composition and the activity coefficients until self-consistency is achieved.

5 Correct Dew-Point Equations

A general algorithm for calculating the dew-point pressure of a mixture at fixed temperature T and specified vapor composition $\{z_i\}$ is as follows.

1. **Given:** T , overall (vapor) composition z_i , and a liquid activity model (e.g. Wilson) with parameters.
2. **Compute pure-component properties at T :** evaluate the saturated vapor pressures $P_i^{\text{sat}}(T)$ using Antoine's equation.
3. **Initialize liquid composition:** choose an initial guess for the unknown liquid composition, e.g.

$$x_i^{(0)} = z_i.$$

4. **Iterative loop:** for iteration $k = 0, 1, 2, \dots$

(a) From the liquid model (Wilson, NRTL, etc.) compute $\gamma_i^{(k)} = \gamma_i(T, \{x_j^{(k)}\})$.

(b) Compute the dew-point pressure using Eq. (7):

$$P_{\text{dew}}^{(k)} = \left[\sum_i \frac{z_i}{\gamma_i^{(k)} P_i^{\text{sat}}(T)} \right]^{-1}.$$

(c) Compute K-values:

$$K_i^{(k)} = \frac{\gamma_i^{(k)} P_i^{\text{sat}}(T)}{P_{\text{dew}}^{(k)}}.$$

(d) Update the liquid composition by material balance:

$$x_i^{(k+1)} = \frac{z_i / K_i^{(k)}}{\sum_j z_j / K_j^{(k)}}.$$

5. **Convergence test:** check whether $|P_{\text{dew}}^{(k+1)} - P_{\text{dew}}^{(k)}|$ and $|x_i^{(k+1)} - x_i^{(k)}|$ are below user-specified tolerances. If not, repeat the iterative loop.

For this assignment, we follow the numerical choices of Example 7.2 and use the given γ_i at the overall composition, which corresponds to taking one non-iterative step with $x_i = z_i$. Even in this simplified case, the correct application of Eq. (7) yields a different dew-point pressure than reported in the text.

6 Numerical Calculation

Using the data from Section 2 and Eq. (7), we compute the dew-point pressure as follows:

$$\frac{z_1}{\gamma_1 P_1^{\text{sat}}} = \frac{0.30}{1.855753 \times 8.2626 \text{ bar}} \approx 0.01957 \text{ bar}^{-1}, \quad (8)$$

$$\frac{z_2}{\gamma_2 P_2^{\text{sat}}} = \frac{0.70}{1.107145 \times 1.7051 \text{ bar}} \approx 0.37080 \text{ bar}^{-1}. \quad (9)$$

Summing these contributions gives

$$\sum_i \frac{z_i}{\gamma_i P_i^{\text{sat}}} \approx 0.01957 + 0.37080 = 0.39037 \text{ bar}^{-1}. \quad (10)$$

Therefore,

$$P_{\text{dew}} = \frac{1}{0.39037} \approx 2.56 \text{ bar}. \quad (11)$$

The example in the text reports $P_{\text{dew}} = 3.948 \text{ bar}$, which is inconsistent with this calculation. The discrepancy confirms the numerical error identified in Section 4.

Once the corrected dew-point pressure is used, the flash calculation procedure in Example 7.2 can be carried out as described in the text. Since the system pressure $P = 5 \text{ bar}$ lies between the bubble-point and dew-point pressures,

$$P_{\text{dew}} \approx 2.56 \text{ bar} < P = 5 \text{ bar} < P_{\text{bubble}} \approx 6.70 \text{ bar},$$

the mixture is indeed two-phase under the specified conditions.

7 Conclusions

In this report we revisited the dew-point portion of Example 7.2 in Chapter 7 and identified two main issues:

1. The reported numerical value $P_{\text{dew}} = 3.948$ bar is not consistent with the dew-point equation and the stated values of γ_i , P_i^{sat} , and z_i . Direct evaluation of the correct dew-point expression (7) yields $P_{\text{dew}} \approx 2.56$ bar.
2. The example does not iterate on the equilibrium liquid composition when computing the dew point, but instead reuses activity coefficients evaluated at the overall composition. A fully consistent algorithm must iterate on both $\{x_i\}$ and P_{dew} while updating the activity coefficients.

Using the corrected dew-point pressure, the flash calculation in Example 7.2 can be performed as intended, and the phase behavior of the ethanol-toluene mixture at $T = 150^\circ\text{C}$ and $P = 5$ bar is confirmed to be two-phase.